Quantitative Investing
Problem Set 5

Ashley Wen (xw564) | Emily Wu (esw63) | Grace Gu (xg325) October 13, 2026

Contents. Part I studies how the factor and corner-portfolio returns vary over the business cycle (Questions (a)–(b)). Part II runs the characteristics-versus-covariances horse race, first on the 25 size and BE/ME portfolios (Questions (c)–(e)) and then on the 25 size and momentum portfolios (Questions (f)–(h)).

Data and Conventions

- **Sample.** Monthly data from 1926-07 to 2026-06 (UMD and the momentum portfolios start 1927-01). The NBER dummy marks 201 recession months across 16 recessions.
- **Returns.** All figures are percent per month. Portfolio returns are in excess of the T-bill; RMRF is already excess, and SMB, HML and UMD are zero-cost long–short returns, so they are used as given.
- **Inference.** Residuals cluster in recession episodes, so the t-statistics in Part I use Newey–West standard errors with 12 lags. Fama–MacBeth standard errors in Part II are the time-series standard deviation of the monthly $\hat{\gamma}_t$ divided by $\sqrt{T}$.
- **Shared code.** All parts load data through `common/data.py` and run Fama–MacBeth through `common/fmb.py`, so lags, samples and beta definitions are identical across Questions (a)–(h).

Part I: Business Cycle Variation

Question (a)

We estimate $R_t = a + b\,D_t + \varepsilon_t$, where $D_t = 1$ in NBER recession months, so a is the non-recession mean and $a + b$ the recession mean.

Table 1: Factor returns in and out of recessions (% per month)

	Non-recession a	Recession $a+b$	Difference b	$t(b)$
RMRF	0.93	−0.47	−1.40	−2.50
SMB	0.19	0.03	−0.16	−0.82
HML	0.40	0.15	−0.25	−0.75
UMD	0.71	0.22	−0.49	−1.00

Newey–West (12 lags) t-statistics. UMD starts 1927-01.

- **What does this tell you?** Only the market is reliably worse in recessions; SMB, HML and UMD give up some return but none of the drops is distinguishable from zero. All three factors still earn a positive average return in recession months, while the market loses money. So the factor premia are not mainly a reward for losing money in recessions, at least not in the way the market premium is.
- **Based on these results, do you believe SMB or HML is more consistent with a risk-based explanation?** SMB is, but only weakly. Its premium all but disappears in recessions, so small stocks fail to pay off exactly when times are bad, which is what a risk story

needs. HML instead keeps a positive return in recessions, in line with Lakonishok, Shleifer and Vishny's (1994) finding that value does not underperform in bad states. With 201 recession months neither difference is significant, so we read this as a lean rather than a verdict.

Question (b)

Table 2: Corner portfolio excess returns in and out of recessions (% per month)

	Non-recession a	Recession $a+b$	Difference b	$t(b)$
Size–BE/ME corners				
Small Growth	0.83	−0.63	−1.46	−1.30
Small Value	1.69	−0.38	−2.07	−2.62
Big Growth	0.90	−0.34	−1.25	−2.21
Big Value	1.24	−0.12	−1.35	−1.51
Size–momentum corners				
Small Losers	0.80	−0.26	−1.06	−1.17
Small Winners	1.84	−0.22	−2.06	−2.98
Big Losers	0.58	−0.83	−1.42	−1.66
Big Winners	1.24	−0.40	−1.64	−2.73
Within-size spreads				
Small Value − Growth	0.87	0.25	−0.62	−1.03
Big Value − Growth	0.34	0.23	−0.11	−0.14
Small Winners − Losers	1.04	0.04	−1.00	−1.29
Big Winners − Losers	0.64	0.39	−0.25	−0.40

Excess returns over the T-bill. Newey–West (12 lags) t-statistics. Momentum portfolios start 1927-01.

- **Do you find stronger cyclical patterns for these extreme portfolios than for the factors?** Yes, but mostly because the corners are long-only portfolios that carry the market's recession losses. Every corner falls by one to two percent per month in recessions, and the high-return corners (Small Value and both Winner portfolios) fall the most. Once we net out the market by taking spreads within a size group, the pattern shrinks back to factor-sized, insignificant drops that stay positive in recessions. The extreme portfolios are more cyclical, but the value and momentum premia themselves are not.

Part II: Characteristics vs. Covariances

25 Size and BE/ME Portfolios

Question (c)

How the PS3 code was extended. We used an LLM to produce most of the PS5 code extension, starting from our PS3 implementation in cells 1(b) and 1(f). The main changes were adding joint regressions on RMRF, SMB and HML to estimate the additional factor betas, generalizing the monthly cross-sectional regression into a function that runs all three specifications, and producing a side-by-side results table. The extension preserved the original FMB standard-error calculation, fixed betas across months, and lagged, time-varying characteristics. The final implementation passes the included verification checks and reproduces the PS3 size and BE/ME estimates to four decimal places when using the original sample and market-only beta. Equation (1) retains the PS3 market-only beta, while equations (2) and (3) use joint three-factor betas, following our stated team convention.

Were the changes correct? Yes. All checks in Cell 7 of our notebook pass:

- Equation (2) equals a single regression of average returns on the fixed betas.
- A numpy re-run of one month confirms which beta each equation uses.
- All equations use the same 1,194 months with 25 portfolios each.
- Every SE equals $sd(\hat{\gamma}_t)/\sqrt{T}$.

Fixed betas, time-varying characteristics. The betas are estimated once and stay fixed across all months of the second pass. Size and BE/ME are lagged and change over time. Checks 1–3 in Cell 7 confirm this structure was preserved.

Table 3: Fama–MacBeth estimates, 25 Size–BE/ME portfolios, full sample (% per month)

	(1) Characteristics	(2) Covariances	(3) Combined
γ_0	1.7547 (4.15)	1.6433 (4.16)	1.4108 (3.53)
γ_M	−0.4777 (−1.34)	−0.9042 (−2.21)	0.2049 (0.52)
γ_{size}	−0.0674 (−2.30)	–	−0.0932 (−1.96)
$\gamma_{B/M}$	0.2327 (4.08)	–	0.4808 (4.39)
γ_{SMB}	–	0.1087 (1.12)	−0.1567 (−0.83)
γ_{HML}	–	0.3752 (3.54)	−0.4656 (−2.05)
Avg. R^2	0.490	0.485	0.573

Average monthly FMB coefficients with t-statistics (testing $\gamma = 0$) in parentheses. 1927-01 to 2026-06, $T = 1194$, $N = 25$. β_M in equation (1) is the market-only (CAPM) beta; in equations (2) and (3) it is the joint FF3 loading, so the two are different numbers. $\ln(size)$ is lagged one month; $\ln(BE/ME)$ is the previous calendar year's value.

Check against PS3. On 1969-07 to 2026-06 ($T = 684$), with the market-only beta re-estimated on that sample, equation (1) gives $\gamma_{size} = -0.0194$ ($t = -0.64$) and $\gamma_{B/M} = 0.1573$ ($t = 2.11$). These match our PS3 results exactly to 4 d.p., and γ_0 and γ_M also match. The full-sample equation (1) estimates differ because the sample period is different.

Implementation qualifications.

- *Market beta in equation (1):* the wording is ambiguous (see the sensitivity check below). With β_M^{FF3}, γ_{size} changes from -0.0674 ($t = -2.30$) to -0.0367 ($t = -1.20$), while $\gamma_{B/M}$ stays at $t > 3$.
- *BE/ME timing:* this follows the PS3/team calendar-year rule, not the Fama–French July–June timing described in Lecture 2.
- *Generated regressors:* the betas are estimated on the full sample, so they contain measurement error and a look-ahead element. The FMB SEs are not adjusted for beta-estimation error.
- *PS3 comparison precision:* the match is confirmed to 4 d.p., which is all PS3 printed.

Sensitivity check: which market beta to use in equation (1). Step 1 of Question (c) says to estimate β_{iM}, β_{iSMB} and β_{iHML} "using full-period OLS regressions on RMRF, SMB, and HML," and the same symbol β_{iM} appears in all three equations. This can be read as one joint FF3 regression whose market loading is used everywhere. However, the same step notes that "β_{iM} was already estimated in Problem Set 3," and we are asked to verify that equation (1) matches the PS3 γ_{size} and $\gamma_{B/M}$. In PS3, β_{iM} came from a market-only regression. The assignment does not settle which reading is intended. The team convention uses the market-only beta β_M^{CAPM} in (1) and the joint FF3 betas (β_M^{FF3}, β_{SMB}, β_{HML}) in (2) and (3). Under the PS3 setup (Cell 7), this reproduces the PS3 estimates and t-statistics to 4 d.p. Table 4 shows what changes with β_M^{FF3} in equation (1);

the sample (1927-01 to 2026-06, $T = 1194$), the characteristics and the FMB standard-error formula are unchanged.

Table 4: Equation (1) under the two market-beta definitions (% per month)

	β_M^{CAPM} (main)	β_M^{FF3} (sensitivity)
γ_0	1.7547 (4.15)	1.5272 (3.76)
γ_M	−0.4777 (−1.34)	−0.5264 (−1.21)
γ_{size}	−0.0674 (−2.30)	−0.0367 (−1.20)
$\gamma_{B/M}$	0.2327 (4.08)	0.2020 (3.57)

t-statistics in parentheses. 1927-01 to 2026-06, $T = 1194$.

Average monthly R^2 moves from 0.4898 to 0.4852, and the two market betas have a cross-sectional correlation of 0.65. The main difference is γ_{size}: its $|t|$ is above 1.96 with β_M^{CAPM} but not with β_M^{FF3}. $\gamma_{B/M}$ has t above 3 in both versions.

Question (d)

Step 1 — Preliminary interpretation. Equation (3) has the highest average monthly cross-sectional R^2 (0.573, versus 0.490 for equation (1) and 0.485 for equation (2)), although this measures in-sample fit and partly reflects the additional regressors, so it does not by itself distinguish between the characteristics and risk explanations. In the combined specification, BE/ME stays strongly significant, and its coefficient roughly doubles relative to equation (1) (0.4808, $t = 4.39$, versus 0.2327, $t = 4.08$). Size remains negative but narrowly misses the 5% level ($t = -1.96$, $p = 0.0508$). By contrast, γ_{HML} turns from positive and significant in equation (2) (0.3752, $t = 3.54$) to negative in equation (3) (-0.4656, $t = -2.05$), and γ_{SMB} is insignificant in both equations ($t = -0.83$ in (3)). Taken together, BE/ME keeps explanatory power after controlling for factor exposures, while the betas lose their expected positive sign, which leans toward a characteristics interpretation; however, the roughly doubled standard errors in equation (3) suggest betas and characteristics are hard to separate, so we treat this conclusion as tentative.

Step 2 — LLM discussion.

- **"What is the strongest economic argument that size and BE/ME are priced because they are characteristics rather than because they co-vary with systematic risk?"** The strongest argument is that investors may systematically misprice firms with certain observable characteristics. For example, investors may overvalue glamorous, low-BE/ME firms and undervalue distressed or neglected high-BE/ME firms, so the later return premium reflects price correction rather than compensation for bearing systematic risk. Under this view, size and BE/ME themselves should predict returns even after controlling for factor betas.
- **"What is the strongest economic argument that size and BE/ME premia are compensation for systematic risk — and what would that imply about the betas on SMB and HML?"** The risk view says small and high-BE/ME firms tend to perform poorly in certain bad economic states, so investors require higher expected returns to hold them. SMB and HML are then factor-mimicking portfolios capturing these common systematic risks. Therefore, firms with higher β_{SMB} or β_{HML} should earn higher average returns, and once those betas are controlled for, size and BE/ME should have little additional explanatory power.
- **"In a Fama–MacBeth horse race between equations (1), (2), and (3), what empirical pattern would cleanly support the characteristics view? What pattern would**

support the risk view?" Suppose equation (1) contains characteristics, equation (2) contains factor betas, and equation (3) includes both:

- Characteristics view: size and BE/ME are significant in (1) and remain significant in (3), while γ_{SMB} and γ_{HML} become weak or insignificant.
- Risk view: γ_{SMB} and γ_{HML} are significant in (2) and remain significant in (3), while γ_{size} and $\gamma_{B/M}$ become weak or insignificant.
- If both remain significant in (3), the evidence is mixed rather than a clean victory for either explanation.

Step 3 — Evaluation.

- **Does the LLM correctly identify the key distinguishing prediction — that if characteristics dominate in equation (3), it is hard to sustain a pure risk story?** The LLM correctly identifies that characteristics remaining significant in equation (3), while beta premia weaken, would challenge a pure SMB/HML risk explanation. However, significance alone is not enough and the signs of the coefficients also matter.
- **Does the LLM correctly note that the horse race is inherently noisy because characteristics and betas are correlated by construction (size sorts produce size beta differences)?** The LLM's answer does not mention that characteristics and betas are strongly correlated because of the portfolio sorts. Our notebook shows correlations of -0.974 between average log size and β_{SMB} and 0.948 between average log BE/ME and β_{HML}, making their separate effects difficult to identify.
- **Does your empirical evidence from equations (1)–(3) more strongly support one side? Be explicit about which coefficients drive your conclusion.** Our full-sample results lean toward the characteristics view, especially for BE/ME: $\gamma_{B/M}$ increases from 0.2327 ($t = 4.08$) in equation (1) to 0.4808 ($t = 4.39$) in equation (3). γ_{size} remains negative but narrowly misses 5% significance ($p = 0.0508$), while γ_{SMB} is insignificant in both equations (2) and (3). γ_{HML} remains significant but reverses from positive to negative (-0.4656, $t = -2.05$), which conflicts with the positive premium predicted by the LLM's risk explanation. The evidence therefore favors characteristics, but the strong correlations make this conclusion tentative.

Question (e)

Table 5: Fama–MacBeth estimates, 25 Size–BE/ME portfolios, 1963-01 onward (% per month)

	(1) Characteristics	(2) Covariances	(3) Combined
γ_0	1.2790 (3.97)	1.1429 (4.42)	1.3700 (3.58)
γ_M	−0.2247 (−0.77)	−0.5261 (−1.73)	−0.2827 (−0.78)
γ_{size}	−0.0437 (−1.52)	–	0.0045 (0.09)
$\gamma_{B/M}$	0.1704 (2.50)	–	0.4876 (3.57)
γ_{SMB}	–	0.1228 (1.10)	0.0527 (0.26)
γ_{HML}	–	0.3364 (3.06)	−0.4949 (−1.87)
Avg. R^2	0.525	0.528	0.593

t-statistics in parentheses. 1963-01 to 2026-06, $T = 762$, $N = 25$, with betas re-estimated on the same months (team conventions). β_M is the CAPM beta in (1) and the joint FF3 loading in (2) and (3).

Table 6: Full sample versus post-1963, 25 Size–BE/ME portfolios

	Equation	Full: Est. (t)	Post-1963: Est. (t)
γ_{size}	(1)	−0.0674 (−2.30)	−0.0437 (−1.52)
$\gamma_{B/M}$	(1)	0.2327 (4.08)	0.1704 (2.50)
γ_{SMB}	(2)	0.1087 (1.12)	0.1228 (1.10)
γ_{HML}	(2)	0.3752 (3.54)	0.3364 (3.06)
γ_{size}	(3)	−0.0932 (−1.96)	0.0045 (0.09)
$\gamma_{B/M}$	(3)	0.4808 (4.39)	0.4876 (3.57)
γ_{SMB}	(3)	−0.1567 (−0.83)	0.0527 (0.26)
γ_{HML}	(3)	−0.4656 (−2.05)	−0.4949 (−1.87)

Full sample: 1927-01 to 2026-06 ($T = 1194$). Post-1963: 1963-01 to 2026-06 ($T = 762$).

- **Does the characteristics-vs.-covariances result change for the post-1963 sample?** For BE/ME, the result is basically the same. In equation (3), $\gamma_{B/M}$ is almost unchanged (0.4876, $t = 3.57$), while γ_{HML} stays negative and is now insignificant at 5% ($t = -1.87$, $p = 0.06$). So the value evidence still leans toward the characteristics view. The size result changes: γ_{size} is no longer significant in equation (1) ($t = -1.52$) and is essentially zero in equation (3) ($t = 0.09$), and γ_{SMB} is insignificant in both samples. After 1963, neither size nor its beta has a statistically significant coefficient in the combined specification, so the size dimension gives no evidence for either side.
- **Do you feel more or less confident in your conclusion from part (d)?** We are more confident about BE/ME, because $\gamma_{B/M}$ in equation (3) has a similar size and stays significant in both samples. We are less confident about size, which was already borderline in part (d) and disappears after 1963. Overall the conclusion is still tentative. With fewer months, the standard errors in equation (3) are larger (e.g. the SE of γ_{HML} is 0.2644 vs. 0.2270), and betas and characteristics are still strongly correlated by construction of the 25 portfolios.
- **Why might the sample period matter for this particular question?**
  - *Data quality:* before 1963, book equity is not from Compustat and covers fewer firms, so BE/ME and the portfolios may be measured less precisely.
  - *In-sample vs. out-of-sample:* the post-1963 period is close to the sample in which the size and value effects were originally documented, so the pre-1963 data works partly as an out-of-sample check.
  - *Changing premia and risk:* the size premium seems to depend on the period. The pre-1963 sample also includes the Great Depression, when distress risk was especially important. If factor loadings change over time, full-sample fixed betas fit some periods worse than others.

25 Size and Momentum Portfolios

Question (f)

Method. We estimate three Fama–MacBeth (FMB) specifications using the 25 size- and momentum-sorted portfolios. Equation (1) includes market beta, log size, and past returns (ret212); equation (2) includes market, SMB, and UMD betas; equation (3) combines characteristics and factor betas. First-stage betas are estimated from time-series regressions of portfolio excess returns on the relevant factors. Following the shared convention, equation (1) uses the CAPM market beta estimated from RMRF alone, while equations (2) and (3) use market, SMB, and UMD betas estimated jointly. Betas remain fixed in the second stage, while $\ln(size)$ and ret212 are lagged one

month and vary over time. Each month, we run the relevant cross-sectional regression and then average the monthly coefficients. FMB standard errors equal the time-series standard deviation of the monthly coefficients divided by the square root of the number of months. All equations use the same 1,186 months from February 1927 to June 2026.

Table 7: Fama–MacBeth estimates, 25 Size–momentum portfolios, full sample (% per month)

	(1) Characteristics	(2) Covariances	(3) Combined
γ_0	2.0220 (5.51)	1.5006 (4.55)	2.2296 (4.91)
γ_M	−0.5967 (−2.21)	−0.6567 (−1.89)	−0.9362 (−2.54)
γ_{size}	−0.1319 (−4.35)	–	−0.0670 (−1.53)
γ_{ret212}	0.0068 (4.49)	–	0.0076 (2.77)
γ_{SMB}	–	0.3616 (3.45)	0.1031 (0.56)
γ_{UMD}	–	0.6179 (4.47)	0.0745 (0.35)
Avg. R^2	0.531	0.543	0.620

Average monthly FMB coefficients with t-statistics in parentheses. Excess returns in percent per month; 1927-02 to 2026-06, $T = 1186$. β_M is the CAPM beta in (1) and the joint RMRF/SMB/UMD loading in (2) and (3). ret212 is the portfolio's past 2–12 month return in percent, lagged one month, so γ_{ret212} is per percentage point of past return: a portfolio whose ret212 is 10 percentage points higher is predicted to earn about 0.07% more per month.

Interpretation. In equation (1), both smaller size and higher past returns predict higher subsequent excess returns. In equation (2), γ_{SMB} and γ_{UMD} are significant when characteristics are excluded. However, in the combined equation (3), γ_{ret212} remains significant ($t = 2.77$), while γ_{SMB} ($t = 0.56$), γ_{UMD} ($t = 0.35$), and γ_{size} ($t = -1.53$) are not. This favors the characteristics interpretation for momentum over a pure explanation based on the measured SMB and UMD exposures, although collinearity and first-stage beta estimation error limit how decisively the two explanations can be separated.

Question (g)

Step 1 — Preliminary interpretation. Before consulting an LLM, our interpretation was that both explanations had some support when tested separately, but the combined model favored momentum characteristics. Size and past returns are significant in equation (1), while γ_{SMB} and γ_{UMD} are significant in equation (2). Once they compete in equation (3), γ_{ret212} remains significant, while γ_{SMB} and γ_{UMD} lose significance. This leans toward the characteristics view, although correlated characteristics and factor exposures prevent a definitive causal conclusion. *We recorded the qualitative view above before the LLM discussion; the numerical values were later updated to match the group conventions.*

Step 2 — LLM interaction. We asked: "For the momentum premium, is the debate between a characteristics explanation and a risk explanation the same as for the value premium? What makes momentum harder to reconcile with conventional risk-based models than value?"

The LLM argued that both value and momentum raise the same broad question of systematic risk versus mispricing. Value has potential links to persistent fundamental risks such as distress, while momentum is based on recent price performance and lacks an equally obvious stable economic risk exposure. It also noted that conventional models have difficulty explaining momentum's return continuation, while behavioral mechanisms such as investor underreaction provide a natural alternative. Momentum also has severe state-dependent crash risk, particularly when past losers rebound sharply after prolonged declines. It also correctly stated that the existence of crash risk alone does not prove that momentum's average premium is compensation for systematic risk.

Step 3 — Evaluation.

- **Does the LLM recognize that momentum has severe, state-dependent crash risk, and correctly explain why the existence of crash risk alone does not establish that momentum's average return is risk compensation?** Partly. It recognizes momentum's state-dependent crashes and correctly avoids the shortcut that a risky strategy must earn its premium because of that risk. However, it does not fully explain why crashes alone are insufficient evidence for a risk premium: risk compensation requires losses to occur systematically in economically bad states, whereas momentum crashes often occur during sharp market rebounds.
- **Does the LLM mention behavioral explanations for momentum, and correctly contrast these with the risk-based story for value?** Yes, though incompletely. It mentions underreaction but not other behavioral mechanisms such as overreaction or investor inattention. Importantly, it does not simply import the value risk story into momentum: it contrasts value's potential links to persistent fundamental risks such as distress with momentum, which lacks an equally obvious stable fundamental-risk exposure.
- **Does your empirical result for momentum differ from your value result in part (d)? Does the LLM's framing help explain the difference?** Our results provide evidence for both sides. The strongest evidence for the risk view appears in equation (2): γ_{UMD} is 0.618% per month, very close to UMD's average return of about 0.60% per month over the same 1,186 months, which is consistent with a risk interpretation when characteristics are excluded. Momentum crashes and time-varying exposures also leave room for conditional risk explanations. However, the characteristics evidence becomes stronger in equation (3). γ_{ret212} remains significant ($t = 2.77$), while γ_{UMD} falls from 0.618% to 0.075% per month. The equation (3) UMD risk price is far below UMD's average return of 0.60% per month and is itself insignificant ($t = 0.35$), suggesting that UMD beta no longer commands the return predicted by the simple traded-factor risk story once past-return characteristics are included. γ_{SMB} is also insignificant. The beta-only model is not a complete successful risk model either: its intercept remains large and significant, while γ_M is negative. Overall, the evidence leans toward characteristics, but it does not prove that momentum is purely behavioral because characteristics and factor exposures are highly correlated and more complicated risk models could still matter.

  The value result is similar but not identical. In both cases, the characteristic retains explanatory power in the combined specification while the corresponding factor-beta evidence becomes less supportive of a conventional positive risk-premium story. For value, $\gamma_{B/M}$ remains strongly significant in equation (3) (0.4808, $t = 4.39$), while γ_{SMB} is insignificant ($t = -0.83$). γ_{HML} remains significant but becomes negative (-0.4656, $t = -2.05$), which is inconsistent with the positive premium predicted by the standard HML risk story. For momentum, γ_{ret212} remains significant ($t = 2.77$), while γ_{SMB} ($t = 0.56$) and γ_{UMD} ($t = 0.35$) are insignificant, although γ_M is significantly negative ($t = -2.54$). Thus, neither model has a perfectly clean risk-pricing pattern, and momentum does not provide a stronger or better-identified characteristics result. The correlation between β_{UMD} and the momentum characteristic is very high, just as β_{HML} and BE/ME are highly correlated for value, and the full-sample γ_{ret212} t-statistic of 2.77 is lower than the 4.39 of $\gamma_{B/M}$.

  The LLM's framing helps explain the difference only partly. Momentum has less obvious fundamental-risk foundations than value, making its premium harder to reconcile with conventional risk models, so it is less surprising that UMD beta is not priced once the past-return characteristic is included.

Question (h)

Method. We repeat the three specifications using return months from January 1963 onward and re-estimate all first-stage betas within the restricted sample. Lagged characteristics are constructed before the sample is restricted, so January 1963 uses information available in December 1962. The restricted sample contains 762 months from January 1963 to June 2026.

Table 8: Fama–MacBeth estimates, 25 Size–momentum portfolios, 1963-01 onward (% per month)

	(1) Characteristics	(2) Covariances	(3) Combined
γ_0	1.4933 (5.11)	1.1499 (4.83)	1.2531 (3.82)
γ_M	−0.5506 (−2.18)	−0.4002 (−1.43)	−0.9542 (−2.55)
γ_{size}	−0.0578 (−2.20)	–	0.0102 (0.19)
γ_{ret212}	0.0075 (4.98)	–	0.0080 (4.13)
γ_{SMB}	–	0.1734 (1.47)	0.1577 (0.68)
γ_{UMD}	–	0.6659 (4.35)	−0.0427 (−0.20)
Avg. R^2	0.574	0.591	0.658

t-statistics in parentheses. 1963-01 to 2026-06, $T = 762$. All first-stage betas are re-estimated on this sample. Units and beta definitions as in Table 7.

Table 9: Equation (3), full sample versus post-1963 t-statistics, 25 Size–momentum portfolios

	Full sample	Post-1963
γ_M	−2.54	−2.55
γ_{size}	−1.53	0.19
γ_{ret212}	2.77	4.13
γ_{SMB}	0.56	0.68
γ_{UMD}	0.35	−0.20

- **Does the characteristics-vs.-covariances result change for momentum in the post-1963 sample?** The main conclusion does not change after 1963. In the combined equation (3), γ_{ret212} remains strongly significant ($t = 4.13$), while γ_{SMB} ($t = 0.68$) and γ_{UMD} ($t = -0.20$) remain insignificant. The γ_{ret212} coefficient changes only modestly from 0.00759 to 0.00804, while its FMB standard error falls from 0.00274 to 0.00195. The size evidence weakens: in equation (1), γ_{size} falls from −0.1319 ($t = -4.35$) to −0.0578 ($t = -2.20$), and in equation (2) γ_{SMB} loses significance after 1963 ($t = 3.45$ versus $t = 1.47$). In equation (3), the γ_{size} t-statistic moves from −1.53 in the full sample to 0.19 post-1963, so the size effect largely disappears. Thus, the more robust result is the explanatory power of the past-return characteristic rather than size. As in the full sample, these results favor a characteristics interpretation relative to the measured SMB and UMD exposures, although they do not eliminate more complicated risk-based explanations.
- **Compare this to your findings for value in part (e).** Both value and momentum show fairly stable characteristics-versus-covariances conclusions across the full and post-1963 samples. For value, $\gamma_{B/M}$ in equation (3) is almost unchanged, moving from 0.4808 ($t = 4.39$) to 0.4876 ($t = 3.57$). γ_{HML} remains negative, while the absolute value of its t-statistic falls from 2.05 to 1.87. The size characteristic becomes essentially zero after 1963. For momentum, γ_{ret212} also remains stable in magnitude, moving from 0.00759 ($t = 2.77$) to 0.00804 ($t = 4.13$), while γ_{UMD} remains insignificant in both samples. The post-1963 equation (3) UMD risk price (−0.043% per month) is again far below UMD's average return over the same months (0.62%).

- **Which premium appears more stable across sample periods, and what might that tell you?** It is difficult to conclude that one premium is clearly more stable overall. In equation (3), both characteristic coefficients are nearly unchanged across the two samples. In equation (1), however, BE/ME weakens more than ret212. The safest conclusion is that the characteristics-over-betas result survives the sample restriction for both value and momentum, while the size dimension is less stable. Because the post-1963 sample is contained within the full sample rather than being an independent out-of-sample period, this comparison should not be interpreted as a strong independent stability test.

Statistical limitations. This is a historical FMB horse race rather than a real-time trading strategy. Full-period betas are generated regressors estimated using future observations relative to early sample months. FMB standard errors use the standard time-series SD divided by $\sqrt{T}$ and are not Shanken-corrected or HAC-adjusted. Characteristics and estimated factor exposures are also highly correlated, so changes in significance do not by themselves establish causality.